ONLINE STORE SALES ANALYSIS

Project Report
Reporting Period: 2025
Tools: Microsoft Excel
Project Type: Sales Data Analysis

1. Project Overview

This project analyzes an online store's sales data for the year 2025 using Microsoft Excel. The dataset contains 1,000 unique orders across four regions and four product categories. The main objective is to evaluate sales performance, identify revenue and profit patterns, compare regional performance, examine product contributions, and understand customer payment preferences.
The analysis uses data cleaning techniques, calculated fields, PivotTables, PivotCharts, and an interactive-style dashboard to transform raw sales data into meaningful business insights.

2. Data Cleaning and Preparation

The original dataset contained 1,001 records, including one duplicate order. The data preparation process involved the following steps:

- Removed one duplicate record based on the Order_ID field, resulting in 1,000 unique orders.
- Identified one missing Quantity value and two missing Payment_Method values.
- Replaced the two missing Payment_Method values with "Unknown".
- Standardized Region values using uppercase formatting and removed unnecessary spaces from text fields.
- Cleaned product names by removing extra spaces and applying consistent capitalization.
- Created a Data_Quality_Flag column to identify records with missing quantity values.
- Added calculated columns for Revenue, Total Cost, Gross Profit, and Profit Margin.

The cleaned dataset was then used for further analysis and dashboard development.

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3. Key Performance Indicators (KPIs)

The following KPIs were calculated to evaluate the overall sales and profitability performance:

KPI	RESULT
TOTAL ORDERS	1,000
TOTAL REVENUE	$149,501.95
TOTAL COST	$81,669.00
GROSS PROFIT	$67,832.95
AVERAGE ORDER VALUE	$149.50
GROSS PROFIT MARGIN	45.37%

These indicators provide an overview of the store's sales volume, revenue generation, costs, and profitability.

4. Business Insights

4.1 Revenue by Product Category

The Home category generated the highest revenue, contributing approximately 60.8% of total revenue. Electronics ranked second with approximately 24.0%.
This indicates that sales performance is heavily concentrated in the Home category.

4.2 Revenue by Region

The NORTH region generated the highest revenue at $57,174.55 (38.2%), followed by SOUTH at $46,841.15 (31.3%).
The EAST region recorded the lowest revenue at $7,067.85 (4.7%). This difference suggests an opportunity for further investigation into regional sales performance.

4.3 Product Performance

Office Chair was the leading product, generating $72,231.75, or approximately 48.3% of total revenue.
The three highest-revenue products—Office Chair, USB-C Hub, and Keyboard—accounted for approximately 68.3% of total revenue.
This indicates a high concentration of revenue among a small number of products.

4.4 Monthly Revenue Trend

A monthly revenue analysis was developed to examine changes in sales performance throughout 2025. A line chart was used to visualize monthly revenue trends.

4.5 Payment Method Distribution

Cash represented the largest share of orders (26.9%), followed by Credit Card (25.6%), PayPal (24.3%), and Bank Transfer (23.2%).
The distribution was relatively balanced across the four payment methods, with no single method dominating order volume.
These findings describe patterns in the available dataset. Further analysis would be required to identify the underlying causes of regional or product-level differences.

5. Tools and Technical Skills Used

Primary Tool: Microsoft Excel
Technical Skills Applied:

- Data Cleaning and Data Validation
- Duplicate Detection and Removal
- Handling Missing Values
- Excel Formulas and Functions (IF, IFERROR, SUM, AVERAGE, COUNTA, TRIM, PROPER, UPPER)
- Calculated Columns and Financial Metrics
- PivotTables and PivotCharts

- KPI Development
- Data Visualization
- Dashboard Design
- Business Data Interpretation

These techniques were used to organize, analyze, and present sales data in a structured and meaningful way.

6. Limitations and Recommendations

6.1 Data Limitations

- One order had a missing Quantity value. Since the actual quantity could not be verified, it was not estimated and its revenue and cost contributions were calculated as zero.
- Two orders had missing Payment Method values, which were categorized as "Unknown".
- The dataset covers only one year (2025), limiting the ability to identify long-term trends or seasonal patterns.
- The analysis focuses on revenue and gross profit. It does not include operating expenses, marketing costs, or customer acquisition costs.

6.2 Recommendations

Based on the findings, the following actions are recommended:

- Investigate the strong revenue concentration in the home category and assess opportunities to diversify sales across other categories.
- Examine the low revenue recorded in the EAST region to identify potential differences in demand, customer activity, or sales coverage.
- Monitor the performance and inventory availability of high-revenue products, particularly Office Chair.
- Continue monitoring monthly revenue to identify seasonal patterns and changes in sales performance.
- Improve data collection procedures to reduce missing values and strengthen reporting accuracy.
- Conduct further analysis using customer-level and order-level data to better understand purchasing behavior.

7. Conclusion

This project demonstrates how Microsoft Excel can be used to clean, analyze, and visualize business data. Through data preparation, KPI calculations, PivotTables, PivotCharts, and dashboard development, the project provides an overview of sales performance and highlights key revenue and profitability patterns.
The analysis also identifies areas for further investigation and demonstrates practical skills relevant to entry-level data analysis Excel reporting roles.